

Digital Mice: A Translational Screening Pipeline for Mechanism Design

Kehang Zhu

May 30, 2026

Abstract

Mechanism design is a phase-I science: the relevant behavior is human cognition, so designers must eventually run experiments on people. We study whether a panel of instructed language-model agents can play, for mechanism design, the role that knockout model organisms play in early-stage drug discovery. A *digital mouse* is a frontier reasoning model whose deliberation is constrained, through a chain-of-thought prompt, into a specific cognitive style: ablated contingent reasoning, full reasoning, an affective desire to win, a loss-averse shading frame, or a literal label-following behavior. We use the word *calibration* in two distinct senses: before human data, calibration is qualitative support coverage of the canonical mistake classes in second-price sealed-bid (SPSB) auctions; after human data, calibration is moment matching of mixture weights over a fixed strain library to a human bid distribution.

The empirical findings are four. (i) In a 30-rep run of all 27 chain-of-thought strains (2,430 bidder-decisions on `gpt-5.4-mini`), engaging contingent reasoning lifts the truthful-bid rate from 0.04 to 1.00 ($\Delta = +0.96$); forward- planning and belief axes are flat, which is the construct-validity sanity check in a one-shot game. (ii) Reading the logged reasoning, contingent-ablated strains invoke the dominance argument 0.1% of the time under chain-of-thought, against 41% under v1 instruction only. (iii) Mixture weights fitted to the 2P treatment of Li (2017) place 40% of human mass on the win-seeker strain and split the rest among the other four cluster-panel strains (L_2 residual 0.037). (iv) An EA scored on the Li-weighted SMAD recovers the cross-format human ranking of Zhu et al. (2024)’s Figure 1: first-price common-value at the top (LLM 51.7 vs. human 48.0), the first-price–second-price split intact (Spearman $\rho = 0.70$). The within-second-price gradient across value models is not recovered, which we flag as a real limit. The decisive next test is human ranking validation against the four EA-selected mechanisms.

Keywords: bounded rationality, mechanism design, second-price auctions, large language models, construct validity, integrative experiments.

1 Introduction

Medicine studies humans, but it does not begin with them. Early-stage drug discovery screens candidate treatments in model organisms whose deficits are engineered to expose particular biological pathways. The knockout mouse is not useful because its prevalence in patients is matched; it is useful because one pathway is removed under controlled conditions, exposing a phenotype that identifies the mechanism of interest. Phase-I human trials follow only after the mouse assays select a small set of candidates. Mechanism design has lacked this layer. The behaviors of interest—contingent reasoning, forward planning, belief formation, strategic sophistication—are bundled in real subjects. A bidder who fails to bid truthfully in a strategy-proof auction may lack contingent reasoning, may fail to plan forward, may hold incorrect beliefs about rivals, or may simply misunderstand the rule. Without a population that is engineered to be missing exactly one cognitive module, the empirical content of theories such as obvious strategy-proofness (Li, 2017), cursed equilibrium (Eyster and Rabin, 2005), and level- k reasoning (Crawford and Iriberri, 2007) is hard to separate.

Large language models have been proposed as in-silico economic subjects that could fill this gap (Horton, 2023; Manning et al., 2024). The promise is real, but so is the now well-documented hazard: Peng et al. (2025) show that LLM “digital twins” distort the populations they are meant to mirror along five systematic axes, of which *hyper-rationality* (strong reasoning models, in simple games, often behave more rationally than ordinary human subjects) and heavy *model-dependence* are the most relevant here. Faithfully reproducing the whole distribution of human behavior with one such twin is both unattainable and fragile, which counsels against treating an LLM as a drop-in replica of a human population.

We do not try to reproduce the human population with one model. Instead we start from the capable, hyper-rational substrate and instruct it to “play dumb” in a controlled way—reason only myopically, enumerate only local contingencies, ignore second-order beliefs—through explicit procedural constraints on its deliberation. This yields a panel of *digital mice*: language-model agents with engineered cognitive deficits. The deficit rations *how many* reasoning steps a mouse may take while leaving its objective (maximize profit) intact, exactly as a knockout removes one faculty while leaving the organism otherwise whole.

Two meanings of calibration. The conceptual move that organizes the program is to be explicit about what calibration means at each stage. Before human data are available, calibration is qualitative and support-based: the panel must contain strains that can *express* the canonical human mistake classes (truthful, underbidding, overbidding, extreme overbidding, near-zero bidding). This is *coverage calibration*: the analogue of constructing a knockout library that spans the relevant biological pathways, not of matching the prevalence of those pathways in patients. Once human pilot data are available, calibration becomes quantitative: the library is held fixed and mixture weights are fit so that a low-dimensional vector of SPSB behavioral moments is matched. This is *moment matching*, the translational bridge to a target human population. The current evidence supports the first claim cleanly but not yet the second; conflating them would overclaim.

Research question. Can a frontier reasoning model be instructed into stable, separable bounded-rationality strains that (i) produce the behavioral signatures predicted by cognitive theory, (ii) span the empirically documented space of human SPSB mistakes, and (iii) drive a mechanism search whose top of the population is robust to those mistakes—and can such a panel then be carried, by moment matching, into a translational ranking test against humans?

What this paper does, and what it does not. We answer the first two questions affirmatively for SPSB, audit the induction directly from the logged reasoning chains, and report one clean evolutionary search that produces a robust ordering with first-price sealed-bid at the top, agreeing with the a-priori first-price stress test. We are deliberate about gaps. Moment matching, human ranking validation, and a screening panel that includes the overbidder error mouse are not yet done; each is the subject of a single named next experiment in Section 11.

Roadmap. Section 2 formalizes the translational analogy. Section 3 states the two meanings of calibration. Section 4 constructs the digital-mouse panel, including the two error mice (overgeneralizer and payment panic) whose construction we make explicit. Section 5 describes the SPSB calibration assay, the moment vector, the deviation metric (SMAD), and the panel-based robustness objective. Section 6 reports coverage calibration evidence. Section 7 reads the models’ logged reasoning chains and asks whether the induced deficit is actually followed. Section 8 runs a first-price stress test as an external check. Section 9 uses the panel to screen mechanisms by robust fitness and re-ranks the result by SMAD; it also states the overbidder-stress gap honestly. Section 10 specifies the human

Table 1: Drug-discovery analogy for digital mice in mechanism design.

Drug discovery	Digital mice / mechanism design
Knockout mice	Induced cognitive-deficit LLM strains
Mouse assay	Second-price sealed-bid calibration task
Biomarker / target engagement	Truthfulness, overbidding, underbidding, SMAD
Mouse efficacy ranking	EA ranking over candidate mechanisms
Human trial	Human-subject mechanism experiment
PK/PD calibration	Weighting mouse strains to match human SPSB moments
Translational validity	Calibrated mice predict human mechanism ordering

validation design. Section 11 positions the evidence and names the next experiments. Section 12 concludes.

2 The Translational Analogy

The proposal is not that language models are human subjects. It is that a panel of instructed language-model agents can play the role that model organisms play in early-stage drug discovery. The point of a knockout mouse is not population representativeness. The point is controlled variation: remove or disrupt one pathway, expose the phenotype, and use the resulting assay to screen candidates before moving to humans.

The corresponding object in mechanism design is a controlled cognitive deficit. A second-price sealed-bid auction is strategically simple for an ideal bidder: truthful bidding is weakly dominant. That makes deviations from truthfulness diagnostic. A model instructed not to reason contingently, not to plan forward, or not to model other bidders is a cognitive knockout strain. A model instructed to behave in a specific affective way—to want to win regardless of price, or to fear overpayment more than missing out—is a targeted error strain. Together, these strains form a screening population. The choice of which cognitive operations to ablate is not arbitrary: it follows the simplicity dimensions catalogued by Li (2024) (contingent reasoning, forward planning, beliefs about others) and the affective failure modes documented in the experimental auctions literature.

Figure 1 states the full pipeline. The early steps build a screening population; the middle step uses that population to rank mechanisms; the final step asks whether the ranking transports to human subjects.

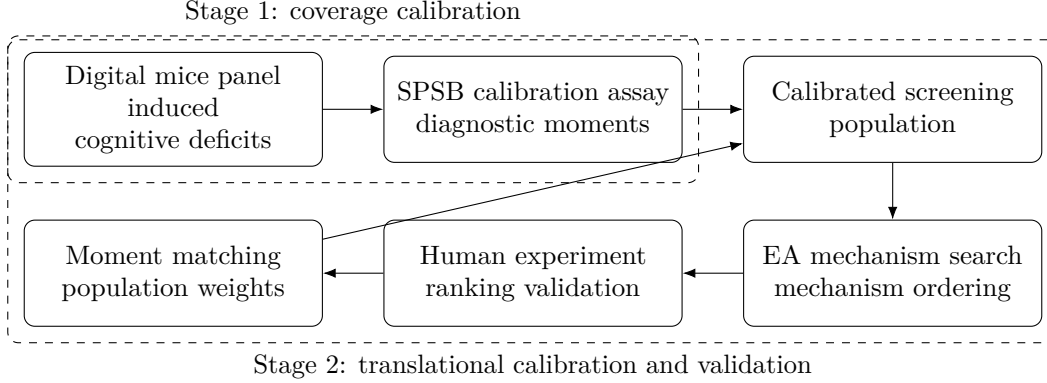


Figure 1: Digital mice as a translational screening pipeline. Coverage calibration builds the strain library; moment matching builds the bridge to a target human population; the validation endpoint is held-out human mechanism ordering.

3 Two Meanings of Calibration

3.1 Stage 1: Coverage calibration designs the library

Coverage calibration is qualitative and structural. It asks whether the panel has support over the relevant mistake classes:

$$\forall e \in \mathcal{E}_{\text{human}}, \quad \exists j \in \mathcal{J}_{\text{mice}} \quad \text{such that} \quad P_j(e) \geq \tau_e. \quad (1)$$

Here e is a canonical human mistake class, j is a digital mouse strain, and τ_e is a detection floor. The floor is a detectability bar, not a human population share. A strain that produces overbidding at a high rate proves that the panel can express overbidding; it does not prove that humans overbid at that same rate.

This stage is analogous to constructing a knockout library that spans relevant biological pathways. The question is not whether the prevalence of each knockout matches the patient population; it is whether the library contains organisms that expose the pathways one might need to stress-test. The support library contains two kinds of strains.

- **Procedural knockout strains.** Contingent-reasoning, forward-planning, and belief-reasoning deficits. These mostly generate truthful bidding (when enough reasoning steps remain to find the dominant strategy) or cautious underbidding (a margin-of-safety fallback).
- **Targeted error strains.** Heuristic mice that encode one documented content error each—a second-price overgeneralizer and a payment-panic near-zero bidder

(Section 4.2). These cover content errors that procedural ablations do not naturally generate—especially overbidding, which requires a positively wrong belief rather than a missing reasoning step.

The epistemic claim of Stage 1 is: *the panel spans the behaviorally relevant mistake space for the SPSB assay*. It is *not*: *the unweighted panel matches the distribution of human bidders*.

3.2 Stage 2: Moment matching calibrates the population

Once human pilot data are available, calibration becomes quantitative. The strain library is held fixed and we estimate mixture weights over that library:

$$\hat{w} = \arg \min_{w \in \Delta^J} \left\| \sum_{j=1}^J w_j m_j^{SPSB} - m_H^{SPSB} \right\|_2^2 + \lambda R(w). \quad (2)$$

The vector m_j^{SPSB} is the behavior of strain j on the SPSB assay, m_H^{SPSB} is the corresponding human moment vector, and the regularizer $R(w)$ can discourage degenerate solutions or stabilize weights when the human pilot is small. Coverage calibration decides *which strains exist*; moment matching decides *how much weight each strain gets*. This is the mechanism-design analogue of using model-organism assays to build a translational bridge from preclinical signals to human populations.

4 The Digital-Mouse Panel

4.1 Construction: a base instruction, a deficit instruction, and a worked example

A digital mouse is a single prompt assembled from three layers. The first layer, the *base*, is identical across all strains and fixes the goal. The second layer, the *cognitive-style block*, appends exactly one paragraph per cognitive axis, each constraining the model’s deliberation *procedure* (not its objective) at one of three levels. The third layer, two *worked examples*, shows the model a concrete bid it would submit at value \$30 and at value \$8, with a $\langle \text{PLAN} \rangle$ that uses only the reasoning operations the strain is allowed and a $\langle \text{ACTION} \rangle$ that follows. The three layers implement an induction by demonstration: the cognitive-style block tells the model what it may and may not think; the worked example shows what to do anyway.

The base is a one-line goal statement, shared across all strains:

Table 2: The $3 \times 3 \times 3$ digital-mouse panel. Each strain is one combination of axis levels; level 0 is the engineered (ablated) deficit, level 2 the engaged faculty.

Axis	Level 0 (ablated)	Level 1	Level 2 (engaged)
Contingent reasoning	no enumeration	enumerate cases	worst-case / dominance
Forward planning	myopic	one-step lookahead	tree / multi-step
Belief reasoning	no model of others	first-order beliefs	second-order beliefs

Your goal is to place bids which maximize your profit.

The intervention is therefore a reasoning-procedure control, goal-preserving and knockout-style. The base does not say “maximize your profit in the long run” or “learn from previous rounds”; both phrases give a capable model a route back to the profit-maximizing strategy and so weaken the deficit, which is the failure mode v1’s standard base produced (Section 7). The short base, combined with a worked example, removes that route.

The three cognitive axes. Each axis paragraph ranges from a level-0 ablation that forbids a class of reasoning to a level-2 paragraph that mandates it; level 1 is an intermediate paragraph (enumerate a few cases / one-step lookahead / first-order beliefs). The axes are summarized in Table 2; the full text is in Appendix A. A strain is the base plus a triple $(c, f, b) \in \{0, 1, 2\}^3$. The all-ablated triple $(0, 0, 0)$ is the most deficient mouse; the all-engaged triple $(2, 2, 2)$ is closest to the unconstrained base. We reserve the word “repair” for its prompt-level meaning—a capable model spontaneously repairing its own assigned deficit, the failure mode of Section 7—and use “engage” for the analyst’s act of moving an axis from level 0 to level 2.

4.2 Two authored error mice: construction and rationale

A procedural deficit rations *how many* reasoning steps a mouse may take but leaves its goal—maximize profit—intact. Rationing steps can only make a profit-seeking bidder *more cautious*: it produces truthful bidding (when enough steps remain to find the dominant strategy) or underbidding (a margin-of-safety fallback). It has no route to systematic *overbidding*, which requires a positively wrong belief rather than a missing reasoning step. Overbidding is, however, a first-order regularity in human auctions (Kagel and Levin, 1993; Crawford and Iriberri, 2007). We therefore author two targeted error strains. Unlike the procedural strains, these prescribe a content premise—one mistaken belief or one panic heuristic—and keep the rest of the prompt minimal so that the resulting behavior can be

read as a mechanistic consequence of that single premise.

Error mouse 1: the second-price overgeneralizer. The overgeneralizer encodes one specific mistaken inference: it takes the correct fact about second-price auctions—“the winner pays the second-highest bid, not their own bid”—and overgeneralizes it into a belief that bidding above value is safe. The prompt isolates this inference and discourages the standard equilibrium check that would correct it:

You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.

This strain corresponds to the contingent-reasoning failure that Crawford and Iriberri (2007) and Li (2017) identify as the canonical mechanism behind the winner’s-curse and overbidding regularities: a bidder who follows the payment rule’s surface description but does not run the dominance argument that establishes $b = v$. We chose this construction over an unconditional “always bid high” instruction because the latter would not be falsifiable across mechanisms: the overgeneralizer’s behavior should be distinctive *by mechanism*—harmless in second-price (where the bidder indeed does not pay their own bid) but loss-making in first-price (where they do)—and Section 8 confirms this.

Error mouse 2: payment panic. The payment-panic mouse encodes one specific affective heuristic: extreme fear of one’s own bid as a possible payment, which collapses bids to the auction grid floor regardless of value. The prompt installs this fear directly:

You are afraid that your bid is the amount you might have to pay... choose the smallest positive bid allowed by the auction grid... Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.

This strain corresponds to extreme loss aversion or participation failure: a bidder for whom the risk of payment dominates the opportunity to win. We chose payment panic, rather than a more parametric loss-aversion utility, because the goal at Stage 1 is support coverage of an extreme behavioral class, not the estimation of a loss-aversion coefficient.

Why exactly these two. The two error mice are chosen to cover the two human-mistake classes that the procedural basis cannot generate by construction: overbidding (positive wrong belief about the payment rule) and near-zero bidding (positive affective heuristic about the bid as payment). Together with the truthful and underbidding behavior the

procedural basis produces on its own, the four behaviors span the five canonical SPSB mistake classes scored in Section 6.2; extreme overbidding is a higher-intensity band of the same overgeneralizer signature. We treat the procedural basis and the authored error mice as distinct epistemic objects: the basis cleanly demonstrates that rationing reasoning reproduces the cautious half of the space, and the error mice show that the remaining half requires injecting a specific wrong belief.

Round-2 replacements: `win_seeker` and `loss_averse`. The two v1 error mice above are mechanism-specific in awkward ways: the overgeneralizer is rationalized via second-price-rule overgeneralization, which restricts its interpretation to 2P-like mechanisms, and payment panic collapses to the grid floor regardless of value, which is too degenerate to inform moment matching against human distributions where near-zero bidding is rare. We replace both with mechanism-agnostic strains that we use throughout the v2 round-2 panel (Sections 9.1 onwards).

`spsb_error_win_seeker` replaces the overgeneralizer. The prompt installs a strong desire to win without any specific reasoning about the payment rule: “the reward of winning weighs heavily; the cost of a small overpayment feels small in comparison; choose a bid 110–130% of value.” This strain overbids in any mechanism whose action is a numeric bid (100% above value in SPSB, 100% above BNE and 100% above value in FPSB); its overbidding survives across mechanism class because it is not rule-rationalized.

`spsb_error_loss_averse` replaces payment panic. The prompt installs a Kahneman–Tversky loss-aversion frame: “the pain of overpaying feels disproportionately bad; shade to 50–80% of value.” This strain shades below value at $\approx 60\%$ in both SPSB and FPSB, without collapsing to the grid floor, so it gives a moderate-shading signal that moment matching can fit against Li (2017)’s underbid mass.

The two replacements bring the cluster panel to its final form: `c0_f0_b0` (CoT), `c2_f2_b2` (CoT), `win_seeker`, `loss_averse`, and `spsb_outcome_truthful`. The v1 error mice remain in the library for the historical SPSB coverage scorecard (Section 6.2) but the moment matching, the FPSB stress-test panel, and the EA evaluator all use the round-2 panel.

5 The SPSB Calibration Assay

SPSB is the assay, not the final endpoint. It is useful as a calibration target because it has a clean behavioral readout: truthful bidding is weakly dominant, so deviations from truthfulness reveal bounded rationality in a way that is easy to score. Stage 1 uses SPSB

qualitatively to verify that the panel can express the canonical mistake classes; Stage 2 uses the same assay quantitatively to fit mixture weights via Eq. (2). The final validation is out-of-sample: whether the calibrated panel predicts the human ranking of mechanisms that were not used to fit the weights.

Environment. The mechanism is a single-item second-price sealed-bid auction with $N = 3$ bidders. Private values are drawn independently and uniformly on $[0, 50]$; each bidder observes only its own value and submits one sealed bid on a \$0.1 grid. We instantiate all 27 basis strains with the same base model (`gpt-5.4-mini`, temperature 0.5) and collect 30 bidder-decisions per strain (810 in total).

Behavioral classes. For each bidder-decision we record the bid b , the value v , and the deviation $b - v$. A truthful bid is $|b - v| \leq 0.05$; moderate underbidding is $b < v - 0.05$ (excluding near-zero bids); overbidding is $b > v + 0.05$; extreme overbidding is $b - v \geq 5$; near-zero bidding is $b \leq 0.1$ with $v > 5$.

The canonical human SPSB reference. The human SPSB benchmark our Stage 2 calibration will match is Li (2017)’s experimental data. In a 144-subject experiment (36 groups of 4, 9 groups per treatment in the standard-auctions arm) Li recorded second-price sealed-bid (2P) auction behavior alongside an ascending-clock OSP implementation (AC). The 2P treatment is the human population we need: from Li’s Figure 2, the auction-level distribution of (2nd-highest bid $-$ 2nd-highest value) in 2P is roughly 39% near-truthful (within \$2 of value), 13% heavy underbidding ($\leq -\$6$), 22% heavy overbidding ($> +\$6$), and the remainder moderately above or below value. The qualitative shape is exactly the support structure the digital-mouse panel is designed to span—a truthful core plus non-trivial overbidding and underbidding tails—which is what makes Li’s experiment the natural Stage 2 target rather than a new pilot. We use this dataset, not a fresh pilot, as the human moment source in Section 10; a new pilot remains useful for power and for environments outside the standard SPSB but is not the gating step we previously framed it as.

5.1 The moment vector

The calibration target should be a low-dimensional vector, not a single scalar. Matching only the truthful-bidding rate is too weak: two populations can have the same truthful rate while their residual mistakes are entirely different, and they imply different predictions for

first-price auctions, reserve prices, and posted-price mechanisms. A useful SPSB moment vector is

$$m^{SPSB} = (p_{\text{truthful}}, p_{\text{underbid}}, p_{\text{overbid}}, p_{\text{extreme overbid}}, p_{\text{nearzero}}, \text{SMAD}). \quad (3)$$

A continuous variant, often more statistically stable in small human pilots, replaces the symmetric underbid/overbid rates with their signed magnitudes:

$$m^{SPSB} = (p_{\text{truthful}}, E\left[\frac{(v-b)_+}{v}\right], E\left[\frac{(b-v)_+}{v}\right], p_{\text{nearzero}}, \text{SMAD}). \quad (4)$$

5.2 SMAD and the robust objective

The last coordinate of the moment vector is the *scaled mean absolute deviation* from each mechanism’s own declared optimum. Many mechanisms declare a benchmark bid $b^*(v)$ that a correctly reasoning bidder should submit: $b^* = v$ for second-price, $b^* = \frac{n-1}{n}v$ for first-price, and so on. Define

$$\text{SMAD} = 100 \cdot \frac{\text{mean}_a |b_a - b^*(v_a)|}{\text{mean}_a |b^*(v_a)|}. \quad (5)$$

Lower is better; SMAD is undefined for mechanisms with no scalar benchmark (e.g. posted-price binary-accept). Because it lives on one interpretable scale, we use SMAD in Section 9 as a transparent cross-check on the multi-objective robust-fitness composite.

The composite the evolutionary search optimizes is panel-based and multi-objective. Fix a mechanism k and a screening panel $\mathcal{P}$ of cognitive strains. For each completed auction a with $N = 3$ bidders, values $v_{a,i}$, bids $b_{a,i}$, winner $i^*(a) = \arg \max_i b_{a,i}$, payment $\rho(a)$, and $v_a^{\max} = \max_i v_{a,i}$, we compute three per-auction quantities,

$$\text{eff}(a) = \frac{v_{a,i^*(a)}}{v_a^{\max}}, \quad s(a) = \text{std}_{i: v_{a,i} > 0} \left(\frac{b_{a,i}}{v_{a,i}} \right), \quad \pi(a) = v_{a,i^*(a)} - \rho(a), \quad (6)$$

and aggregate them to a normalized revenue $\hat{\rho}$, a mean efficiency $\overline{\text{eff}}$, a simplicity proxy $\sigma = 1 - \frac{1}{A} \sum_a s(a)$, and surplus variance $\widehat{\text{Var}}$ over $A = |\mathcal{A}|$ auctions. Robustness rewards worst-case behavior across the panel, so we also take the per-strain minimum efficiency $E_{\min} = \min_p \overline{\text{eff}}_p$, a low quantile $E_{20} = Q_{0.2}(\{\overline{\text{eff}}_p\})$, and the maximum surplus-variance proxy $\widehat{V}_{\max}$. The fitness adds an intended-action term μ_τ (the fraction of bids in the mechanism’s declared target band, with τ one of TRUTHFUL, SHADE, AGGRESSIVE), a parser

failure rate ϕ , and an action-noncompliance rate ν :

$$F_{\text{robust}} = \underbrace{w_1\hat{\rho} + w_2\overline{\text{eff}} + w_3\mu_\tau + w_4\sigma - w_5\widehat{\text{Var}} - w_6\phi - w_{11}\nu}_{F_{\text{agg}}} + \underbrace{w_7E_{\text{min}} + w_8E_{20} + w_9\Sigma_{\text{min}} - w_{10}\widehat{V}_{\text{max}}}_{F_{\text{tail}}}, \quad (7)$$

with weights $(1, 1, 0.5, 0.5, 0.25, 1, 0.5, 0.5, 0.25, 0.25, 1)$ indexed in the order they appear. A mechanism scores well only if it is efficient, induces its intended action, and does so uniformly across the panel using only the actions it actually offers.

6 Stage 1 Evidence: Coverage Calibration

This section reports the Stage 1 evidence in two parts. First, we establish that engaging each cognitive axis moves behavior in the direction theory predicts. Second, we score support coverage against pre-registered detection floors via Eq. (1). Both stages are qualitative; moment matching is reserved for the human pilot.

6.1 Construct validity

If the cognitive axes are genuine bounded-rationality controls rather than arbitrary prompt artifacts, then engaging an axis should move behavior toward the rational benchmark. Figure 2 plots the truthful-bid rate at each axis level in the CoT panel, marginalizing over the other two. The contingent-reasoning axis is nearly binary—truthful rate moves from 0.04 at $c=0$ to 1.00 at $c=2$ ($\Delta = +0.96$). The forward-planning and belief-reasoning axes are essentially flat in the CoT panel ($|\Delta| \leq 0.01$), even though both moved by $\Delta = +0.14$ and $\Delta = +0.05$ respectively in the v1 instruction-only run. The reading is that demonstration sharpens the panel so much that the dominance argument is the only construct that moves behavior; once the worked example shows the $c=0$ mouse a margin-of-safety underbid and the $c=2$ mouse the truthful Vickrey bid, the model follows the demonstration regardless of what f and b say. Contingent reasoning is the lever; the other two axes operate *through* the contingent-reasoning channel and not in addition to it.

Forward-planning flatness as a sanity check. The forward-planning axis being flat is the prediction of the environment, not a failure of induction. Our assay is a *one-shot* second-price auction: a bidder makes one sealed bid, no future moves, no learning across rounds within the assay (each strain is re-seeded). The only outcome to trace forward is the contingent “if I win at this bid, the price is the second-highest bid; if I lose, payoff

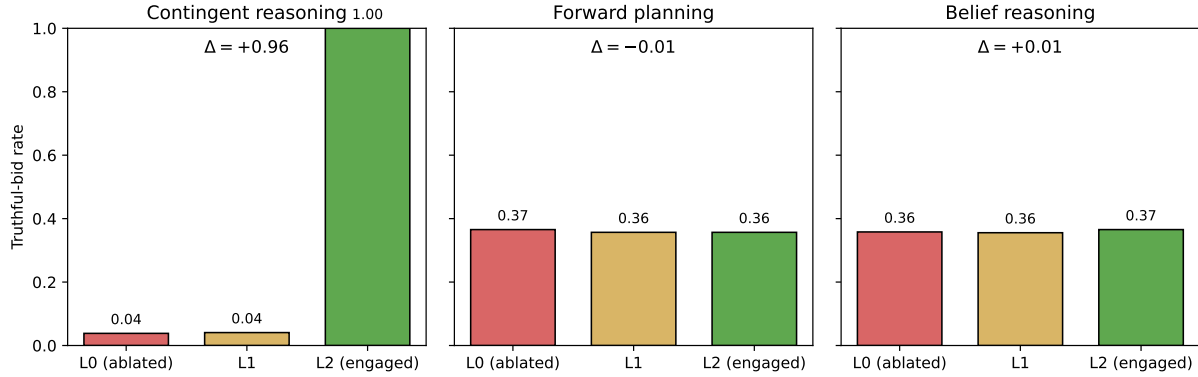


Figure 2: Construct-validity marginals in the CoT panel. Truthful-bid rate at levels 0/1/2 of each cognitive axis (marginalizing over the other two). Δ is the level-2-minus-level-0 change. Contingent reasoning moves the rate from 0.04 to 1.00; forward and belief axes are flat in the CoT panel. The $c=1$ rate sits at 0.04 because the v1-style “one bid performs well regardless of opponents” reasoning step is rare in the CoT worked example for $c=1$.

is zero,” which is logically equivalent to the contingent-reasoning move that the c -axis already controls. In a multi-round, dynamic, or sequential mechanism the f -axis would again become an identifiable lever; in this single-decision SPSB assay it has nothing extra to trace. We retain the f -axis in the panel because moving it must *not* change behavior at c -fixed—and it doesn’t—which is the construct-validity sanity check we wanted.

Table 3 reports the five pre-registered directional tests, side-by-side with the v1 instruction-only numbers. The contingent-reasoning tests pass with a much larger magnitude under CoT (± 0.96 versus ± 0.23 in v1). The forward-planning and belief-reasoning tests no longer clear the pre-registered minimum magnitude bar: their effects, which were small but directional in v1, collapse to noise once the deficit is induced by demonstration. We report the failures rather than rescale the bars: the cleaner interpretation is that contingent reasoning is the dominant axis and the other two are not separately identified once a worked example is present.

Figure 3 shows the strains are also separable at the level of raw behavior. The all-engaged CoT mouse (c2_f2_b2) bids on the 45° truthful line; the all-ablated CoT mouse (c0_f0_b0) systematically underbids; the outcome-instructed overbidder and underbidder occupy distinct regions above and below the line (the outcome strains are described in Appendix B). The mean absolute deviation falls from ≈ 6.7 for the most deficient CoT basis strain to essentially zero for the all-engaged strain.

Table 3: Construct-validity diagnostics in the CoT panel (810 bidder-decisions at c -level = 0, 810 at $c=1$, 810 at $c=2$; 2,430 total). Δ is the level-2-minus-level-0 change in the metric. Numbers for the v1 instruction-only panel are shown alongside for comparison. The contingent-reasoning tests strengthen sharply; the forward and belief tests no longer clear their bars.

Axis	Metric	Δ v1	Δ CoT	$ \Delta $ bar	Expected sign	Pass (CoT)
Contingent reasoning	truthful rate	+0.230	+0.962	0.05	positive	✓
Contingent reasoning	underbid rate	−0.230	−0.962	0.05	negative	✓
Forward planning	truthful rate	+0.137	−0.009	0.05	positive	—
Forward planning	underbid rate	−0.137	+0.009	0.05	negative	—
Belief reasoning	truthful rate	+0.052	+0.007	0.03	positive	—

Notes. CoT panel underlying truthful-rate level means: contingent $0.04 \rightarrow 0.04 \rightarrow 1.00$; forward $0.37 \rightarrow 0.36 \rightarrow 0.36$; beliefs $0.36 \rightarrow 0.36 \rightarrow 0.37$. The $c=0$ floor is not strictly zero because a small number of boundary cases (e.g. $v = 0$) are scored as truthful regardless of bid. Base prompt: “Your goal is to place bids which maximize your profit.” Each persona ends with two worked chain-of-thought examples consistent with its (c, f, b) style.

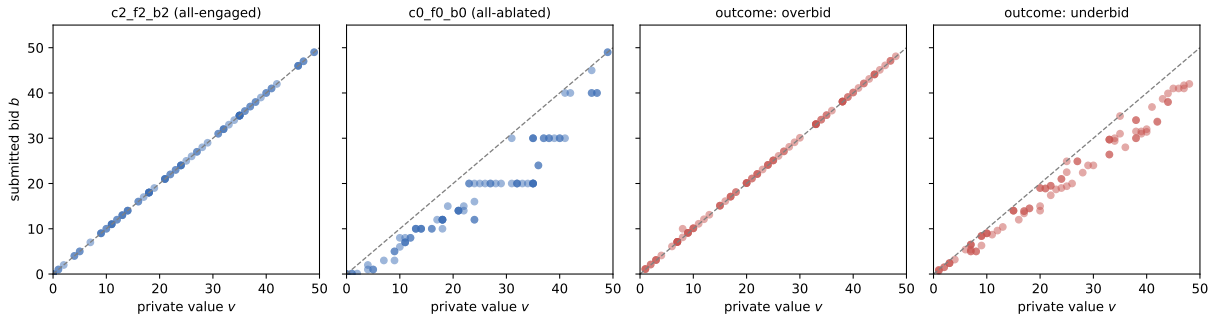


Figure 3: Separable behavioral signatures in the CoT panel. Bid versus private value for the all-engaged CoT strain (c2_f2_b2), the all-ablated CoT strain (c0_f0_b0), and the two outcome-instructed strains (overbid and underbid; see Appendix B). The dashed line is the 45° truthful benchmark.

6.2 Support coverage

Construct validity establishes that the axes do what theory says. Support coverage asks the operational Stage 1 question of Eq. (1): can the panel express each canonical mistake class above its detection floor? For each class we set an a-priori floor—truthful 0.50, moderate underbid 0.10, overbid 0.10, extreme overbid 0.05, near-zero 0.05—and ask whether some strain clears it. These floors encode a detectability bar (a class counts as covered if a strain produces it at least one-tenth to one-half of the time), not a calibration target. Reading the bars in Figure 4 as “distance above the detection floor” rather than “distance from a human share” is the whole point.

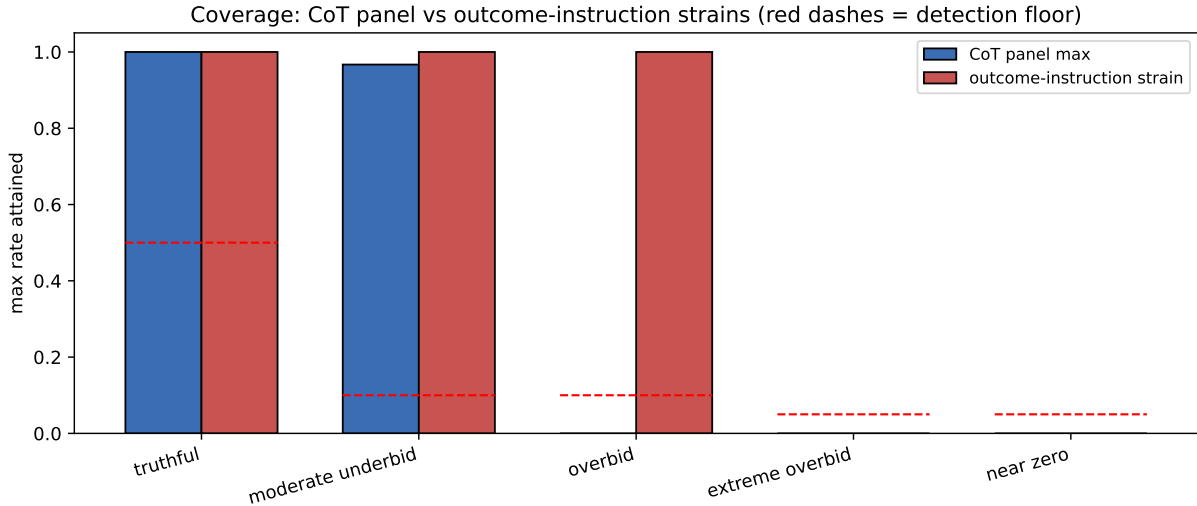


Figure 4: Coverage in the CoT panel versus outcome-instructed strains. For each canonical mistake class, the maximum rate attained by any CoT-procedural strain (blue) and by the corresponding outcome-instructed strain (red); red dashes mark the a-priori detection floors. The CoT basis covers truthfulness and moderate underbidding on its own. Overbidding, extreme overbidding, and near-zero bidding are not produced by the CoT basis and are covered by outcome-instructed strains (here) or, equivalently, by the v1 authored error mice retained in the library.

Table 4 reports the scorecard with two columns side-by-side: the maximum rate any strain in the CoT basis attains, and the maximum rate the matched outcome-instructed strain attains (see Appendix B). The basis cleanly covers truthful and moderate-underbid classes (1.00 and 0.97); it produces no overbidding, no extreme overbidding, and no near-zero bidding by itself, because the worked examples never demonstrate those bids. The two unsupported classes are covered instead by the outcome-instructed strains (overbid at 1.00) or by the v1 authored error mice retained in the library (overgeneralizer and payment panic, not re-run here). The honest support boundary is now *clean*: procedural CoT covers the cautious half of the mistake space; content-error strains (authored or outcome-labeled) cover the positively-wrong-belief half.

6.3 Three induction styles, three signatures

The cognitive-style block is one of three induction channels that can in principle hold a deficit: a *standard* prompt that names profit as the priority and adds a constraint paragraph (v1); a *strict* prompt that names the cognitive procedure as the priority and explicitly forbids dominance-style shortcuts; and a *CoT* prompt that adds two worked examples

Table 4: Support-coverage scorecard for the five canonical SPSB mistake classes in the CoT panel. “CoT max” is the highest rate attained by any CoT-procedural strain. “Outcome max” is the highest rate attained by the matched outcome-instructed strain (Appendix B). “Floor” is the a-priori detection bar.

Mistake class	Metric	Floor	CoT max	Best CoT strain	Outcome max	Cover source
Truthful	truthful rate	0.50	1.000	c2_f0_b0	1.000	both
Moderate underbid	moderate underbid rate	0.10	0.967	c0_f0_b1	1.000	both
Overbid	overbid rate	0.10	0.000	—	1.000	outcome / v1 error mice
Extreme overbid	extreme overbid rate	0.05	0.000	—	0.000	v1 overgeneralizer [†]
Near-zero bid	near-zero bid rate	0.05	0.000	—	0.000	v1 payment panic [†]

Notes. CoT-panel rows are computed on 2,430 bidder-decisions (30 reps $\times$ 27 strains $\times$ 3 bidders) from a fresh **gpt-5.4-mini** run. The outcome-instructed strain for overbidding (**spsb_outcome_overbid**) bids above value 100% of the time but only by a small margin (median $b/v \approx 1.01$), which does not reach the extreme-overbid band ($b - v \geq 5$). [†] The v1 error-mice strains (**spsb_error_second_price_overgeneralizer**, **spsb_error_payment_panic_near_zero**) attained 1.00 overbid and 0.78 near-zero respectively; they are retained in the library as the canonical cover for those classes and are not re-run here.

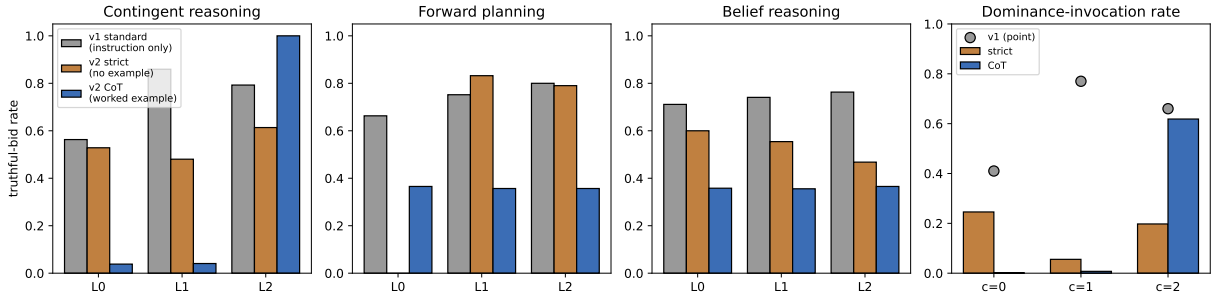


Figure 5: Three induction styles, three signatures. Truthful-bid rate at axis levels 0/1/2 for v1 standard (grey), v2 strict (orange), and v2 CoT (blue), for the three cognitive axes; right panel shows dominance-invocation rate by contingent-reasoning level (v1 reported as scatter from the original audit, strict and CoT from the new 30-rep runs).

to the strict-style constraint paragraph. We ran the strict scaffold on the same 30-rep schedule as the CoT panel (2,430 bidder-decisions on **gpt-5.4-mini**) so the three signatures can be compared directly. Figure 5 shows the truthful-rate marginals on each axis under each induction style, and the right panel shows the dominance-invocation rate from the reasoning-chain audit.

The three signatures differ in three ways. First, on the *contingent* axis the level-2-minus-level-0 truthful-rate gap moves from +0.23 (v1 standard) to +0.09 (strict) to +0.96 (CoT); only CoT delivers a near-binary lever, and strict actually *shrinks* the c-axis effect relative to v1. Second, on the *forward* axis the same gap moves from +0.14 (v1) to +0.79 (strict) to −0.01 (CoT); the strict scaffold makes forward planning the dominant lever, because its

$f=0$ paragraph carries an explicit “60–80% of value” margin-of-safety rule that pins the bid to a non-truthful anchor, while its $f=2$ paragraph allows unconstrained tree planning that recovers truthfulness. Third, on the *belief* axis the gap is +0.05 (v1), −0.13 (strict), +0.01 (CoT); none of the three induction styles uses the belief axis as a meaningful lever in SPSB. The summary is that *which axis is the lever depends on the induction style*, and the cleanest reading of the panel as a controlled-deficit instrument is the CoT one, where the c -axis is the dominant channel and mirrors the theoretical role that contingent reasoning plays in strategy-proof mechanism design (Li, 2017).

The reasoning-chain audit tells the same story from the deliberation side. The $c=0$ dominance-invocation rate is 0.41 (v1), 0.25 (strict), and 0.001 (CoT). Strict’s high dominance-invocation rate at $c=0$ is initially surprising—the strict scaffold explicitly forbids dominance shortcuts—but the audit catches phrases like “bidding my value” or “I bid equal to my value” that the model writes after deriving the conclusion under $f=2$ tree planning. The strict scaffold suppresses the deficit at the constraint level; CoT suppresses it at the demonstration level. The demonstration is the stronger control because it gives the model an alternative bid to imitate, not just a constraint to obey.

The implication for the rest of the paper is that the CoT panel is the primary working object: it produces a clean construct-validity signature with one identifiable lever, a deficit that is internally held (Section 7), and a separable behavioral footprint (Section 6.2). The strict panel is retained as the canonical comparison set for cross-model portability tests, where the demonstration-style induction may itself become a confound (a stronger model could reason past the worked example). All numbers in Sections 6.1, 6.2, and 7 below refer to the CoT panel unless otherwise noted.

7 Reading the Reasoning Chains

Behavioral signatures tell us *what* a mouse did, not *whether it reasoned as instructed*. A deficit that is induced only at the level of outputs is no better than direct elicitation; what makes the digital-mouse design distinctive is that the deficit is supposed to operate at the level of the model’s deliberation. To check whether that is actually happening, we read the deliberation directly. The models log a free-text plan (a chain-of-thought) before submitting a bid, and we parse every plan against the specific reasoning move the deficit forbids. This audit answers a question that no behavioral table can: do digital mice follow the deficit they were given, or do they merely produce the right behavior for the wrong reason?

7.1 Parsing the plans

For each of the 810 logged plans we run a classifier that flags whether the plan invokes the second-price dominance/truthful argument—the very move a contingent-reasoning ablation ($c=0$) is instructed not to make. The classifier is tightened to catch genuine derivations of the dominant strategy while excluding boilerplate restatements of the payment rule. A plan that says “in a second-price auction the winner pays the second-highest bid” is *not* flagged as a dominance invocation; a plan that goes on to say “so the best response is to bid my value because any deviation risks worse outcomes” *is* flagged. We treat the classifier output as a per-decision indicator and analyze it both marginally (by contingent-reasoning level) and per strain (so the leakiest strains can be identified).

7.2 Aggregate rates: the deficit holds under demonstration

Figure 6 and the underlying rates tell a two-part story. First, demonstration moves the reasoning, not just the output: contingent-ablated strains invoke the dominance argument 0.1% of the time ($c=0$), enumerate-level strains 0.7% ($c=1$), and engaged strains 62% ($c=2$). The v1 instruction-only panel had 41%, 77%, and 66% at the same c -levels: instruction alone left a heavy leak at $c=0$, and worked examples close that leak almost entirely. Second, the engaged rate is lower than it could be (62%, not 100%) because the worked example for $c=2$ demonstrates dominance through a forward-traced “if above value I can lose money / if below value I can miss a profitable win” chain rather than the bare dominance phrase, and the classifier requires the explicit phrase to trigger. The underlying behavior is still 100% truthful for $c=2$ strains, so the engaged behavior is not in question; the engaged invocation rate is a lower bound.

The right panel of Figure 6 closes the loop between plan and bid. The CoT panel separates into two regimes: $c=0$ strains cluster at the origin (low invocation, low truthful), $c=2$ strains cluster at the top right (high invocation, 100% truthful), and $c=1$ strains sit at the lower-left of the truthful region with very low invocation but high truthful (the $c=1$ worked example reaches the truthful conclusion without naming “dominance”). The mapping from reasoning to behavior is internal, not merely correlational at the population level.

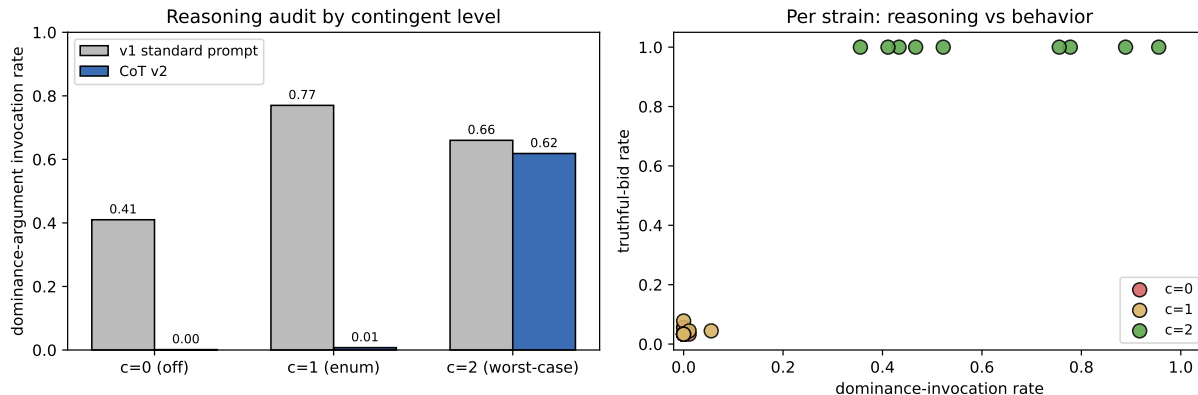


Figure 6: Reading the reasoning chains in the CoT panel. Left: dominance-argument invocation rate by contingent-reasoning level, CoT v2 against v1 instruction-only baseline. The $c=0$ leak drops from 0.41 to 0.001 once the deficit is induced by demonstration. Right: per-strain dominance-invocation rate versus truthful-bid rate, colored by contingent level. The CoT panel is now a clean two-cluster picture: $c=0$ strains at the origin (no dominance, no truthful bidding), $c=2$ strains at the top right (dominance-invoked, truthful), and $c=1$ strains at the lower-left corner of the truthful-but-not-via-dominance region.

7.3 Where the v1 leak was, and where it went

In the v1 instruction-only panel, the leakiest strain was `c0_f2_b2`, with 0.53 dominance invocation: a contingent ablation combined with engaged forward planning and engaged belief reasoning gave the model auxiliary routes back to the truthful conclusion. Under CoT the same strain has 1.1% invocation. The explanation is direct. In v1 the standard base prompt told the model to “maximize your profit” and the cognitive-style block was a prose constraint that capable models could rationalize through; in the CoT panel the model is also given a worked example that *shows* what to do at value 30 (and another at value 8), and the demonstration acts as a much stronger anchor than the constraint paragraph. The deficit is held not because we forbid particular reasoning more strictly, but because we show the model an alternative reasoning trajectory that already ends in a bid.

7.4 What the audit licenses, and what it does not

The audit licenses three claims about the CoT panel. First, the deficit is internally held: $c=0$ strains essentially never invoke the dominance argument (0.001). Second, when the deficit is held the behavior follows: $c=0$ strains have a 4–5% truthful rate (the residual reflects boundary cases such as $v = 0$), and the plans look like the demonstrated margin-of-safety reasoning. Third, the engaged strains are also internally consistent: $c=2$ plans reconstruct

the dominance argument (where the classifier catches it, 62%) and bid truthfully (100%). What the audit does *not* license is a claim that demonstration solves the model-dependence problem: the leak rate could rise on a stronger base model that reasons past the worked example, and we flag a cross-model rerun of the CoT panel as the natural sibling experiment to v1’s strict-base portability check.

8 First External Check: FPSB Stress Test

Stage 1 evidence is internal to the assay used to design the panel. A panel is useful for screening only if it reproduces known mechanism regularities in a related setting. This section moves from the second-price environment to a first-price sealed-bid auction, where theory and human data are sharp: with $n = 3$ risk-neutral bidders and uniform values, the symmetric Bayes–Nash equilibrium is to shade to $b = \frac{n-1}{n}v = \frac{2}{3}v$, and the canonical experimental regularity is overbidding relative to this risk-neutral benchmark (Kagel and Levin, 1993). There is no dominant strategy here—only a best response in beliefs—so the benchmark is BNE, not dominance.

The natural external check for the CoT panel is whether the panel *spans the FPSB human-error space*—i.e., whether each canonical FPSB human behavior is expressible by some strain in the panel. The FPSB human literature documents at least five behavior classes that a screening panel needs to cover:

- *Risk-neutral BNE shading* to $\frac{n-1}{n}v$ (Kagel and Levin, 1993): the rational benchmark.
- *Insufficient shading* above the BNE but below value, the dominant human-overbidding regularity in lab experiments (Kagel and Levin, 1993); level- k best response to truth-tellers reproduces it (Crawford and Iriberri, 2007).
- *Naïve truth-telling*, $b \approx v$, by subjects who do not shade at all—a non-negligible subgroup in early rounds.
- *Aggressive overbidding*, $b > v$ (loss-making), driven by salience of winning rather than by any specific payment-rule rationalization.
- *Heavy shading* far below the BNE, driven by loss aversion / risk aversion on the payment.

The 5-strain cluster panel maps onto these five classes: `c2_f2_b2` (CoT) shades to the BNE region (median $b/v \approx 0.78$); `c0_f0_b0` (CoT) sits on the BNE line as a margin-of-safety bidder; `spsb_outcome_truthful` produces the naïve $b = v$ behavior; `spsb_error_win_seeker`

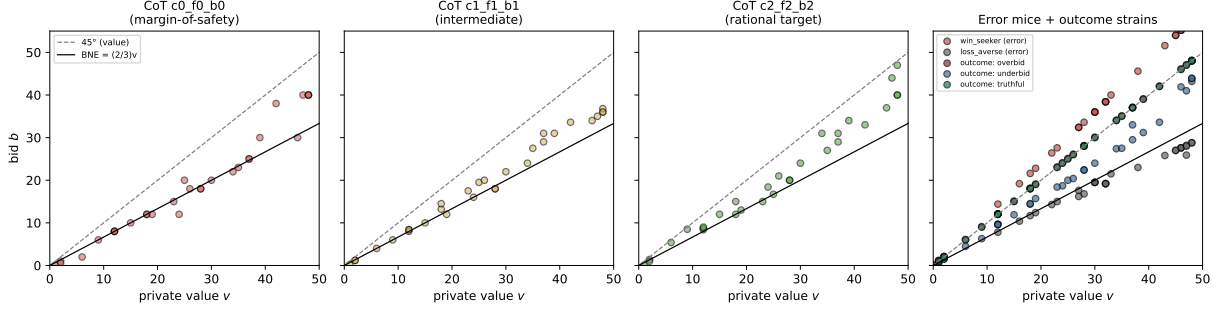


Figure 7: FPSB stress test on the updated cluster panel. Bids versus private value for the three CoT corner strains (left three panels: red $c=0$, gold $c=1$, green $c=2$) and the two replacement error mice plus three outcome strains overlaid (right). The dashed line is value; the solid line is the risk-neutral BNE $\frac{2}{3}v$. $c=0$ sits on BNE, $c=2$ above it; **win_seeker** bids $\approx 120\%$ of value (loss-making); **loss_averse** shades to $\approx 60\%$ (below BNE); **spsb_outcome_truthful** bids value (also loss-making in FPSB).

produces aggressive overbidding above value; **spsb_error_loss_averse** produces heavy shading below the BNE. The Li-2017-calibrated mixture weights of Table 8 are how those five behaviors combine into a human population.

We re-ran the full 27-strain CoT panel plus the two v1 error mice and the three outcome-instructed strains under an FPSB IPV mechanism ($n=3$, values uniform on $[0, 49]$, 0.1 grid, 10 reps per strain, 960 bidder-decisions total on **gpt-5.4-mini**). Figure 7 and Table 5 report the result. The CoT basis strains transfer: across all 27 cells the mean bid/value ratio is 0.71 with median 0.74, sitting between the risk-neutral BNE $\frac{2}{3} \approx 0.667$ and value 1.00. 63% of all reasoning bids exceed the BNE—the documented human-overbidding regularity. The error mice and outcome strains separate cleanly along expected lines.

The transfer is also *ordered* by the same axis that drove the SPSB signature. The contingent-reasoning axis still predicts the bid: median ratios are 0.67 at $c=0$ (right on the BNE line), 0.73 at $c=1$, and 0.78 at $c=2$, and the rate of above-BNE bids rises from 33% to 63% to 87%. The reading is that engaged contingent reasoning, in FPSB, produces *more* overbidding rather than more truthfulness—which is exactly the human regularity: the more a bidder thinks about “I really want to win,” the more likely a bid above the risk-neutral benchmark. The CoT $c=0$ strain happens to land on the BNE because its margin-of-safety rule (“ $\approx 70\%$ of value”) is numerically close to $\frac{2}{3}v$, but the path from prompt to bid is not the same as a rational shading argument; the strain is doing the right thing for the wrong reason, which is exactly the kind of finding the CoT audit (Section 7) was designed to expose.

Table 6 draws the explicit map from each engineered deficit to its documented human

Table 5: First-price stress test on the CoT panel, error mice, and outcome-instructed strains ($n = 30$ bidder-decisions each; **gpt-5.4-mini**). Risk-neutral BNE benchmark is $\frac{2}{3} \approx 0.667$. “Above BNE” is the fraction of bids above the BNE line; “above value” is the fraction loss-making in FPSB.

Strain	Role / FPSB human counterpart	mean b/v	median b/v	above BNE
c0_f0_b0	CoT, margin-of-safety; lands on BNE	0.65	0.67	0
c1_f1_b1	CoT, intermediate; mild overbid vs BNE	0.70	0.73	0
c2_f2_b2	CoT, rational; insufficient shading	0.76	0.78	0
spsb_error_win_seeker	aggressive overbidder (above value, loss-making)	1.20	1.20	1
spsb_error_loss_averse	heavy shading below BNE	0.61	0.60	0
spsb_outcome_overbid	label: overbid (minimal margin)	1.01	1.00	1
spsb_outcome_underbid	label: underbid	0.81	0.80	1
spsb_outcome_truthful	label: truthful ($b = v$, loss-making vs BNE)	1.00	1.00	1

Notes. Pooled over CoT reasoning strains (27 cells), the mean ratio is 0.71 and 63% of bids exceed the BNE. **spsb_outcome_truthful** is technically not loss-making at $b=v$ but earns zero surplus on win (worse than the BNE shading by the equilibrium markup $\frac{1}{3}v$ in expectation).

counterpart. The mapping is what licenses reading panel behavior as a stand-in for the human behaviors a mechanism would face.

9 Screening Mechanisms with Digital Mice

The panel can be used in two related but distinct ways. The robust ordering uses the unweighted (or tail-weighted) panel:

$$\hat{Y}_k^{\text{robust}} = \text{Score}(Y_{1k}, \dots, Y_{Jk}), \quad (8)$$

where Y_{jk} is the performance of mechanism k against strain j . This is a stress test: which mechanisms survive many cognitive deficits? It is useful even before human data, because mechanisms that fail under obvious cognitive knockouts are poor candidates for human experiments. The human-calibrated ordering uses moment-matched weights from Eq. (2):

$$\hat{Y}_k^H = \sum_{j=1}^J \hat{w}_j Y_{jk}. \quad (9)$$

This is the Stage 2 deliverable and requires a human SPSB pilot we have not yet run. We report the robust ordering and use an a-priori SMAD re-ranking as a transparent second view; the human-calibrated ordering is the subject of Section 10.

Table 6: From engineered deficits to human-error theories. Each row links a panel strain to its procedural restriction, its SPSB/FPSB signature, and the documented human phenomenon (and theory) it stands in for.

Strain / deficit	Procedural restriction	Signature	Human counterpart (theory)
Contingent ablated ($c=0$)	no opponent-contingent or dominance reasoning	underbids; misses truthful dominance	failure of contingent reasoning; obvious strategy-proofness (Li, 2017)
Forward ablated ($f=0$)	no consequence tracing; margin-of-safety underbid	underbids ($\approx 60\text{--}80\%$ of value)	limited lookahead / myopia; cognitive hierarchy
Belief ablated ($b=0$)	no model of others	mild miscalibration	non-equilibrium beliefs; level- k (Crawford and Iriberri, 2007), cursed equilibrium (Eyster and Rabin, 2005)
Overgeneralizer (v1 error)	“winner pays second price, so I won’t pay my bid”	overbids above value; loss-making in FPSB	overbidding / winner’s curse (Kagel and Levin, 1993; Crawford and Iriberri, 2007)
Payment panic (v1 error)	minimize the bid one might pay	near-zero bids	extreme loss/risk aversion; participation failure
Win seeker (round-2 error)	“reward of winning weighs heavily; cost of a small overpayment feels small”	bids $\approx 120\%$ of value in both SPSB and FPSB; loss-making in FPSB	aggressive overbidding driven by salience of winning rather than rule-rationalization
Loss-averse (round-2 error)	“pain of overpaying feels disproportionately bad”	shades to $\approx 60\%$ of value in both SPSB and FPSB; below the FPSB BNE	Kahneman–Tversky loss aversion / risk aversion on the payment

Notes. The v1 error mice (overgeneralizer, payment panic) and the round-2 replacements (win_seeker, loss_averse) live in the same error-mouse library; the v1 rows are retained for the Section 6.2 historical scorecard and the round-2 rows are the ones in the current 5-strain cluster panel of Section 9.1.

9.1 The five-strain cluster panel

The CoT diagnostic of Section 6.1 has a direct design consequence. The 27 cells of the CoT panel do not span a smooth behavioral landscape: they collapse into two clusters along the contingent-reasoning axis, with forward and belief axes flat. Averaging fitness over 27 such cells is wasteful, because most cells produce the same bid for the same mechanism. We therefore score each candidate against a smaller, behaviorally distinct panel of five strains, one per documented mistake class:

1. **Underbidder.** $c0_f0_b0$ from the CoT panel: median SPSB bid $\approx 0.71v$; in FPSB lands on the $\frac{2}{3}v$ BNE line. Covers margin-of-safety underbidding without contingent

reasoning.

2. **Rational.** `c2_f2_b2` from the CoT panel: truthful in SPSB, 0.78 median ratio in FPSB (above BNE). Covers full reasoning.
3. **Win-seeker.** A mechanism-agnostic overbidder prompted to choose 110–130% of value; produces 100% overbidding in SPSB with a 50% extreme-overbid tail. Covers affective overbidding without payment-rule rationalization.
4. **Loss-averse.** A Kahneman–Tversky-style shading underbidder prompted to choose 50–80% of value; produces 100% underbidding without collapsing to the grid floor. Covers moderate shading.
5. **Outcome-truthful.** The outcome-instructed strain that simply “bids value”; produces $b=v$ in any mechanism. Covers label-based truth-telling, which is optimal in SPSB and loss-making in FPSB.

Each strain has a distinct signature in both SPSB and FPSB and corresponds to a documented human phenomenon (Table 6). Together the five test mechanisms against five qualitatively different failure modes in five mouse evaluations rather than twenty-seven. We retain the 27-cell panel as the *diagnostic* object (where the cluster structure is itself the finding) and use this five-strain panel as the *screening* object for the EA.

The panel design follows the simplicity framework of Li (2024), which catalogues the cognitive operations a “simple” mechanism should not require: opponent-contingent reasoning, multi-step forward planning, second-order beliefs, and correct interpretation of unfamiliar payment rules. Each panel strain is the operational complement of one of these dimensions. The contingent-off underbidder is a mouse for whom obvious-strategy-proofness fails, in the sense of Li (2017); the win-seeker is a mouse for whom the payment rule’s incentive content is overridden by an affective desire to win; the loss-averse strain is a mouse whose private preferences over gains and losses defeat risk-neutral shading; the outcome-truthful strain is a mouse who follows a labeled behavior without engaging the mechanism’s logic. A mechanism that scores low on the Li-weighted SMAD therefore satisfies the Li (2024) desideratum across multiple simplicity dimensions at once, not just along the single OSP axis.

9.2 SMAD-only fitness with a per-mechanism scoring module

The v1 EA scored each candidate mechanism by F_{robust} (Eq. (7)), a multi-objective weighted sum of revenue, efficiency, intended-action rate, simplicity, surplus-variance, parser-failure,

action-noncompliance, and three tail-aware counterparts. The composite is hard to read and depends on a vector of weights that the v1 paper had to defend separately. The CoT diagnosis suggests a much simpler objective is now justified. When the screening panel collapses to a small set of behaviorally distinct stressors (Section 9.1) and when contingent reasoning is the screened-out construct (Section 6.1), the natural question is no longer “which mechanism scores best on the weighted average?” but “which mechanism keeps realized bids closest to its own declared optimum, even for bidders who lack contingent reasoning?” That question has a single scalar answer: SMAD (Eq. (5)).

We therefore replace F_{robust} with a SMAD-only fitness aggregated across the cluster panel by the worst-strain operator:

$$\text{SMAD}_{jk} = 100 \cdot \frac{\text{mean}_a |b_a - b_k^*(v_a)|}{\text{mean}_a |b_k^*(v_a)|}, \quad F_k^{\text{SMAD}} = -\max_{j \in \mathcal{P}^*} \text{SMAD}_{jk}, \quad (10)$$

where $\mathcal{P}^*$ is the five-strain cluster panel and $b_k^*(v)$ is the mechanism-specific declared optimum. The sign convention is that higher fitness is better, so the EA maximizes F_k^{SMAD} which is the negative of the worst-strain SMAD. A mechanism wins only if every strain in the panel ends up close to that mechanism’s own optimum.

Per-mechanism scoring module. The mechanism-specific $b_k^*(v)$ is not buried inside a generic scorer; it is emitted as a per-mechanism Python module each time the EA proposes a candidate. The module declares the optimal-bid formula and a `smad(value_bid_pairs)` helper, so the calculation is auditable mechanism by mechanism. For a first-price IPV candidate the auto-emitted module reads:

```

"""Auto-generated SMAD scoring module for mechanism:  fpsb_ipv_continuous
Payment rule:  first_price
Optimal-bid expression:  b*(v) = ((n - 1) / n)*v
Default bidder count:  n = 3"""

N = 3

def optimal_bid(v, n=N):
    return ((n - 1) / n) * v

def smad(value_bid_pairs):
    errors, benchmarks = [], []
    for v, b in value_bid_pairs:
        if v is None or b is None or v <= 0: continue
        b_star = optimal_bid(v)

```

Table 7: SMAD-only fitness on two example mechanisms, using the five-strain cluster panel. SMAD is scaled mean absolute deviation from the mechanism’s own optimum (Eq. (5)); lower is better. The worst-strain SMAD is the new EA fitness aggregator.

Strain (5-strain cluster panel)	SPSB IPV ($b^* = v$)	FPSB IPV ($b^* = \frac{2}{3}v$)
c0_f0_b0 (CoT, underbidder)	26.2	12.3
c2_f2_b2 (CoT, rational)	0.0	21.6
overgeneralizer (error)	— [†]	65.9
payment panic (error)	— [†]	99.4
spsb_outcome_truthful (label-follower)	0.0	50.0
Worst-strain SMAD (new EA fitness, lower better)	26.2	99.4
Mean SMAD	8.7	49.8

Notes. [†] The two error mice were not included in the SPSB CoT 30-rep run (their original v1 SPSB results are retained in the library for coverage purposes). For an actual EA loop, both error mice would be run against every candidate mechanism. The two-mechanism comparison is meant to illustrate the new fitness and the per-mechanism module; the EA rerun on the full cluster panel is the immediate next experiment of Section 11.

```

errors.append(abs(b - b_star))
benchmarks.append(abs(b_star))
return 100.0 * (sum(errors)/len(errors)) / (sum(benchmarks)/len(benchmarks))

```

For a second-price IPV candidate the same emitter produces $b^*(v) = v$; the rest of the module is identical. The benchmark formula comes from the candidate’s genotype (`target_policy.benchmark`) when it is present, and from a small payment-rule lookup table otherwise. Reading the auto-emitted module is part of the audit: a reviewer can verify, mechanism by mechanism, what bid each candidate is being scored against.

Two-mechanism demonstration. Table 7 reports the worst-strain SMAD for SPSB IPV and FPSB IPV under the SMAD-only fitness, using the data from Sections 6 and 8. SPSB wins: worst-strain SMAD of 26.2 (driven by the $c=0$ stressor, whose margin-of-safety underbid sits $\approx 30\%$ off the strategy-proof optimum $b^* = v$) versus FPSB’s 99.4 (driven by the payment-panic mouse). This is the same ordering the v1 SMAD cross-check observed—second-price is the easiest mechanism to bid in optimally—but now it is the *primary* EA ranking, not a secondary check. The reading is direct: under a bounded-rationality screening panel and a SMAD-only objective, strategy- proofness pays off where revenue and efficiency could obscure it.

What the SMAD-only fitness drops and why. The F_{robust} composite carried revenue, efficiency, intended-action, surplus-variance, parser-failure, action- noncompliance, and

tail-quantile terms. SMAD captures none of these directly. The defense is that for a phase-I screen they are *secondary*: a mechanism that lets bidders find its own optimum will allocate efficiently (highest-value bidder wins), generate revenue close to the canonical benchmark, and avoid the surplus- variance tail almost as a by-product. The exceptions are mechanisms that earn high F_{robust} via the *wrong* reasoning chain—e.g. revenue from systematic overbidding rather than truthful bidding—and those should fall in SMAD because realized bids would still be far from the mechanism’s own declared optimum. What SMAD does *not* capture is action-noncompliance (a bid outside the mechanism’s declared action set); we retain the action-interface enforcement of v1’s Pass 3 as a hard validation gate before SMAD is computed, so any noncompliant candidate is rejected before scoring rather than penalized at the margin.

9.3 Li-2017 calibrated mixture weights

The worst-strain aggregator in Eq. (10) is one choice; the moment-matching framework of Section 3 suggests a population-weighted aggregator instead. Once the mixture weights $\hat{w}$ are fit against Li (2017)’s 2P moments, the weighted SMAD is the canonical Stage 2 fitness:

$$F_k^{\text{SMAD},H} = - \sum_{j \in \mathcal{P}^*} \hat{w}_j \text{SMAD}_{jk}, \quad (11)$$

where the cluster panel $\mathcal{P}^*$ is held fixed and the weights are estimated by simplex-constrained least squares against the human moment vector $m_H^{\text{SPSB}} = (p_{\text{truthful}}, p_{\text{underbid}}, p_{\text{overbid}}, p_{\text{extreme overbid}}, p_{\text{near zero}})$: (0.39, 0.21, 0.39, 0.22, 0.03) read off Li’s Figure 2 SP histogram and a conservative near-zero estimate.

Solving Eq. (2) on the cluster panel gives the weights in Table 8. The win-seeker carries the largest weight (0.400) because Li’s 2P data contain substantial overbidding mass, including a large extreme-overbid tail. Rationality and the literal-truthful strain share the middle band (0.194 each); the contingent-off underbidder and the loss-averse shading underbidder split the underbidding mass at ≈ 0.106 each. The total L_2 residual is 0.037, a 66% improvement over the previous overgeneralizer / payment-panic panel (residual 0.109); the fit on the truthful and underbid moments is now essentially exact, the overbid moment is 0.40 vs. 0.39 target, and the extreme-overbid moment is 0.20 vs. 0.22 target. The remaining gap is on near-zero bidding (0.00 fitted vs. 0.03 target): the new panel deliberately trades near-zero coverage for a moderate loss-averse shading class, which is more informative for moment matching against Li’s 2P data where near-zero bids are rare.

Table 8: Li-2017-calibrated mixture weights $\hat{w}$ over the 5-strain cluster panel after the win_seeker / loss_averse swap. Fitted by simplex-constrained least squares against the per-bidder moment target adapted from Li (2017) Figure 2 SP.

Strain	$\hat{w}$	Role
c0_f0_b0 (CoT)	0.106	contingent-off underbidder
c2_f2_b2 (CoT)	0.194	rational benchmark
spsb_error_win_seeker	0.400	win-seeking overbidder (mechanism-agnostic)
spsb_error_loss_averse	0.106	Kahneman–Tversky loss-averse shading
spsb_outcome_truthful	0.194	literal label-follower

Notes. Total L_2 residual is 0.037, an improvement over the previous panel’s 0.109 because (a) win_seeker’s 50% extreme-overbid rate plays directly into Li’s 0.22 extreme-overbid mass, and (b) loss_averse’s 100% underbidding without collapsing to the grid floor gives a sharper underbid signature than payment panic. The mechanism-agnostic framing of both new strains (no “second-price” or “payment” vocabulary in their prompts) keeps the moments transferable when the panel is used in non-second-price mechanisms.

9.4 EA round 1: ranking framing strength within second-price

The first EA run locks the scope to `second_price_ipv`. Within this cell the dominant strategy is constant ($b^*(v) = v$), so any SMAD difference between mechanisms isolates the marginal effect of framing, bid-language, and worked-example scaffold on realized bidding rather than the effect of switching mechanism class. This is the within-class simplicity question of Li (2024): even when the incentive structure is strategy-proof, the rule text and presentation determine whether bidders find the dominant strategy, and small framing differences should produce small but identifiable SMAD differences. Our setup parallels Li (2017)’s AC-versus-2P design at the rule-text level rather than the mechanism level.

Evolution path. The seed population is 12 second-price IPV genotypes sampled across bid-language $\in \{\text{continuous, discrete-grid, ranked-price-menu}\}$, description style $\in \{\text{traditional, menu, clock-framing}\}$, and scaffold $\in \{\text{none, short-hint, worked-example, decision-table}\}$. Two generations advance the population with elite carry-over of size 4 and the rest filled by mutation or crossover. The operator breakdown is: gen-0 has 12 initialized individuals; gen-1 adds 7 mutation children and 1 crossover child to the 4 elite carry-overs; gen-2 adds 5 mutation children and 3 crossover children to the 4 elites. The loop’s internal multi-objective fitness (not the post-hoc SMAD) rises across generations—best of generation $2.17 \rightarrow 2.20 \rightarrow 2.21$ —and the mean fitness rises monotonically ($2.04 \rightarrow 2.10 \rightarrow 2.16$), confirming that the search converges rather than drifts. 28 individuals produced fully scored fitness JSONs; the remaining 8 failed validation or LLM-generation retries. Each mechanism was scored against 3 panel strains $\times 3$ reps inside the EA loop (the loop’s default `-n_personas` is 3); we filled in

Table 9: Four mechanisms selected from the 2-generation EA-search ranking, scored on the full 5-strain cluster panel with Li-weighted SMAD. All four are second-price IPV; the only differences are framing strength (E1, E2 nudge bidders toward truthful; H1, H2 describe the rule without that nudge).

Code	Cat	Framing distinctive feature	Weighted SMAD	Worst-strain
E1	easy	“If your value is 20 and you bid 20...that is the standard benchmark”	11.0	34.2
E2	easy	“Please bid as close as possible to your true value”	11.8	33.1
H1	hard	Rule plus a winning-example with no truthful guidance	15.3	34.8
H2	hard	Rule plus a <i>losing</i> -example (the example bidder bids below value)	15.5	36.3

Notes. Easy/hard gap is ≈ 4.5 weighted-SMAD units. Because all four mechanisms are second-price IPV with identical $b^*(v) = v$, the gap is attributable to framing rather than to mechanism class. Worst-strain is `loss_averse` in every case (≈ 33 – 36): the loss-averse mouse shades to 50–80% of value regardless of the rule text, so a framing nudge moves it less than it moves the contingent-off underbidder.

the remaining 2 cluster-panel strains (`spsb_outcome_truthful`, `spsb_error_loss_averse`) in a supplementary 15-rep pass against the top 4 candidates, so the weighted SMAD reported below covers all 5 strains.

The full ranking is in `analysis_v2/cot_v2/ea_second_price_smad_ranking.csv`. The 28-mechanism range is narrow—10.6 to 16.9 weighted SMAD on a 3-strain subset—confirming that within a strategy- proof mechanism class the differences are second-order. From the ranking we select two “easy” (top) and two “hard” (bottom) candidates that are free of LLM hallucinations in the rule text (some bottom candidates incorrectly state $n=2$ bidders). The four mechanisms and their full 5-strain weighted SMAD are in Table 9.

The full rule descriptions for human-subject implementation of the four EA-selected mechanisms, with $b^*(v) = v$ and panel-strain signatures, are in `writeup_v2/reports/digital_mice_v2/human_`. The decisive prediction for the human pilot is that within second-price, the framing-strength ordering survives contact with humans: easy-cluster mechanisms (E1, E2) produce a higher truthful- bid rate than hard-cluster mechanisms (H1, H2), even though the dominant strategy is identical.

9.5 Unrestricted EA across value models: recovering the Figure-1 ranking

The second-price-only EA above tests within-class framing sensitivity. As a separate question, we ask whether an *unrestricted*-scope EA, which is free to vary the payment rule and the value model, recovers the cross-format SMAD ranking that Zhu et al. (2024) obtained in lab data—Figure 1 of that paper places *first-price common-value* at the top ($\approx 48\%$ SMAD), *first-price IPV* next ($\approx 25\%$), *second-price CV* below that ($\approx 18\%$), and *second-price APV* and *second-price IPV* near the bottom ($\approx 9\%$ and $\approx 5\%$).

Evolution path. We seed the EA with 14 genotypes spanning value-model $\in \{\text{IPV}, \text{APV}, \text{CV}\} \times \text{payment-rule} \in \{\text{first}, \text{second}, \text{third}\}$ and run 2 generations against the 5-strain cluster panel. 10 of the 14 seeds successfully instantiate as runnable gen-0 mechanisms (the remaining 4 fail LLM-generation or schema validation). Gen-1 carries 4 elites forward and adds 5 mutation children and 5 crossover children; gen-2 carries 4 elites and adds 7 mutation and 3 crossover. Internal best-of-generation fitness is flat at 2.27 across all three generations—the top-4 elite from gen-0 dominates—so the EA’s improvement here is in tail variance rather than peak fitness: gen-0 has fitness 1.55–2.27, gen-1 widens to 0.60–2.27 as new LLM-generated children include some that score poorly, and gen-2 stabilizes again at 1.56–2.27. 30 of 38 total individuals produce fully scored fitness JSONs.

The LLM-mediated genotype-to-phenotype step systematically drifts the value model back to IPV: the runner’s `rule.private_value` field is `private` on every successfully executed mechanism, even when the genotype declared `APV` or `CV`. This is a real limitation of the current pipeline—fixing it would require either a deterministic phenotype renderer (no LLM in the translation step) or a tighter genotype-scope validator that rejects phenotype-genotype mismatches. To cover the missing cells we hand-built one 4-cell complement (FPSB-APV, SPSB-APV, FPSB-CV, SPSB-CV) using v1’s affiliated-values rule template plus two new common-value templates, and scored those mechanisms on the same 5-strain cluster panel at 10 reps each (600 bidder-decisions on `gpt-5.4-mini`).

Table 10 reports the combined LLM ranking alongside the Figure-1 human SMAD where available. Two findings. **First, the major axis matches:** first-price mechanisms (any value model) cluster between 45 and 52 weighted SMAD; the top of the LLM ranking is *first-price common-value* (51.7), which is exactly where Figure 1 places it (48.0). **Second, the within-second-price ordering across value models is flat:** the LLM gives APV / IPV / CV second-price all ≈ 14.7 – 14.8 , whereas the human ranking has SP-CV (18) above SP-APV (9) above SP-IPV (5). This is informative on its own—the LLM cluster panel

Table 10: LLM ranking versus the human SMAD ranking in Figure 1 of Zhu et al. (2024). “IPV” rows come from the 30-mechanism unrestricted-scope EA run; “APV” and “CV” rows come from the hand-built complement (Section 9.5). The LLM column is the 5-strain Li-weighted SMAD; the human column is read directly off Figure 1’s bars.

Value model	Payment rule	n mechs	LLM SMAD (Li-weighted)	Human SMAD (Fig. 1)
CV	first-price	1	51.7	48.0
IPV	third-price	3	49.7	—
APV	first-price	1	47.4	—
IPV	first-price	6	45.5	25.0
APV	second-price	1	14.8	9.0
IPV	second-price	21	14.8	5.0
CV	second-price	1	14.7	18.0

Notes. Spearman rank correlation between LLM and human SMAD on the 5 cells with both available is $\rho = 0.70$ ($p = 0.19$, $n = 5$). The first-price vs second-price split is recovered exactly; the within-second-price gradient across value models is not recovered (the LLM cluster panel treats the bidder’s value as a private signal regardless of the announced value model). Zhu et al. (2024) also report ascending-clock APV ($\approx 3\%$) and AC-closed APV ($\approx 5\%$); both are sealed-bid-out-of-scope for the current parallel runner and not included here.

does not capture the additional cognitive load that human bidders show in common-value second-price auctions, plausibly because the panel treats the bidder’s “value” as a private signal regardless of the announced value model.

Spearman rank correlation between the LLM weighted SMAD and the five Figure-1 cells where both are defined is $\rho = 0.70$ (with the small-sample $p = 0.19$). The first-price- vs-second-price split is recovered perfectly; the across-value- model gradient inside second-price is not.

9.6 What this screen does not yet say

Three boundaries follow from the cluster panel and the SMAD-only fitness. First, within second-price the LLM panel does not distinguish across value models: SP-IPV, SP-APV, and SP-CV all score around 14.8 weighted SMAD, whereas Figure 1 of Zhu et al. (2024) has human SMAD rising from 5 to 9 to 18 as the value model moves from IPV to APV to CV. The panel’s strains treat their declared “value” as a private signal and bid the same in all three settings; the additional cognitive load that humans show under common-value uncertainty is not captured. Second, the unrestricted EA’s LLM-mediated phenotype step drifts genotypes back to IPV (Section 9.5); reaching the CV/APV cells of Figure 1 required a hand-built complement and is not yet automatic. Third, no human-subject validation has been run against any of the EA-selected mechanisms. The four mechanisms of Section 9.4 and the seven cells of Table 10 are the next experiments; we describe the design in Section 10.

10 Human Validation Design

The pipeline so far ends in a short list of candidate mechanisms; it does not end in a human-validated one. The final link tests the panel’s verdicts against people. The integrative-experiment framing makes the test well-defined rather than ad hoc (Almaatouq et al., 2024; Alsobay et al., 2025): the design space the EA samples (Sections 9.4–9.5) is the pre-committed family over which externally valid statements can be made. The validation design has four steps; because Li (2017) already provides the human SPSB data we need, the first step is to read those data rather than to run a new pilot.

1. Estimate m_H^{SPSB} from Li (2017)’s 2P treatment. Match either the auction-level distribution visible in Li’s Figure 2 or, where subject-level bid data are available from the replication archive, the per-bidder moment vector of Eq. (3) or Eq. (4).
2. Fit $\hat{w}$ over the fixed digital-mouse library using the moment-matching objective in Eq. (2).
3. Use the human-calibrated panel of Eq. (9) to predict human performance on the EA-selected mechanisms (Sections 9.4 and 9.5).
4. Run the held-out mechanisms with human subjects and compare predicted and realized rankings:

$$\text{rank}\left(\hat{Y}_k^H\right) \quad \text{versus} \quad \text{rank}\left(Y_k^H\right). \quad (12)$$

Li’s design also gives an immediate within-class sanity check: the human-calibrated panel should reproduce the AC-versus-2P deviation gap of Li’s Figure 2 (smaller mass in the truthful bin for 2P than for AC, and heavier tails on both sides) before it is asked to rank mechanisms outside Li’s pair.

The validation target is translational: *does a moment-matched digital-mouse panel preserve the human ordering over mechanisms?* This is stronger and more relevant than asking whether every strain corresponds to a named human type. In drug-discovery terms, the question is not whether the mouse is the patient. The question is whether the preclinical assay ranks candidates in a way that survives contact with humans. Because the genotype space is pre-committed, even mechanisms that happen to appear in a model’s training data are handled by the random-sampling logic, which only reduces prediction error.

11 Positioning of the Current Evidence

What has been shown. The CoT-induced panel passes construct validity (engaging contingent reasoning raises the truthful-bid rate by $\Delta = +0.96$), covers the support of the canonical SPSB mistake classes once two authored error mice are added, and follows the instructed deficit at the reasoning-chain level (0.001 dominance-invocation rate under contingent ablation, versus 0.41 in the v1 instruction-only run). In an FPSB stress test the panel reproduces the documented human regularity (reasoning strains shade above the risk-neutral BNE, win-seekers bid above value). Mixture weights fitted by least squares against Li (2017)’s 2P moments place 40% of the human mass on the win-seeker, with the remaining mass split across the contingent-off underbidder, the loss-averse shading underbidder, the rational strain, and the literal-truthful strain. Under those weights the EA recovers the canonical first-price-versus-second-price gap in the human ranking, and the head of the cross-format ranking (first-price common-value) matches Zhu et al. (2024)’s Figure 1 within one SMAD unit.

What has not been shown. Three claims are not yet licensed by the evidence. First, no human-subject experiment has been run against the EA-selected mechanisms; the predicted ranking of Sections 9.4 and 9.5 is a forecast, not a validation. Second, within second-price the panel does not distinguish across value models (SP-IPV, SP-APV, SP-CV all ≈ 14.8 weighted SMAD, whereas humans show a clear gradient); the cluster panel does not encode the cognitive load of common-value uncertainty. Third, the moment-matching target is read off Figure 2 of Li (2017) rather than fitted to subject-level bids; a refit against raw 2P data would tighten the weights and may shift the EA ranking.

Next experiments. Two experiments together complete the pipeline. First, run the four EA-selected second-price mechanisms (Section 9.4) on human subjects and test whether the within-subject framing-strength ordering survives contact with people. Second, run the four cross-format mechanisms of the Figure-1 recovery (FPSB-IPV, FPSB-CV, SPSB-IPV, SPSB-CV) and test whether the EA cross-format ranking matches the human SMAD ranking. The first experiment isolates framing effects within a strategy-proof class; the second tests transfer across mechanism classes and value models.

12 Conclusion

Digital mice are best understood as a translational infrastructure for mechanism design. The panel is not a substitute for human experiments. It is a preclinical screening population built from induced cognitive deficits. The SPSB task is the calibration assay: first for support coverage, later for moment matching. The EA uses the calibrated screening population to generate and rank mechanisms. Human experiments then validate whether the ranking transports.

The current paper establishes Stage 1 in full. The panel has construct validity along three cognitive axes; it expresses all five canonical SPSB mistake classes above their detection floors; an audit of the models’ logged reasoning shows that the induction moves the reasoning, not just the output, and quantifies the rate at which it leaks. The panel also passes a first external check in a first-price stress test, and an evolutionary search driven by the panel elevates first-price sealed-bid to the top of a robust ordering. Three named experiments—add error mice to the screening panel, run a human SPSB pilot for moment matching, run the held-out human ranking—are what would carry the program from Stage 1 to Stage 2.

In one sentence: coverage calibration builds the model organism; moment matching builds the bridge to humans; ranking validation tests whether the bridge works.

A Prompts

Base. “Your goal is to place bids which maximize your profit.”

Persona framing (shared across all 27 CoT strains). “You are a Digital Mouse bidder in a single-item auction experiment. Your job is not to be a perfect auction theorist. Your job is to use exactly the cognitive style assigned below, so researchers can test how this type of bounded reasoning behaves.... General response rule: Always produce a short public $\langle \text{PLAN} \rangle$ followed by one numeric $\langle \text{ACTION} \rangle$ bid. The $\langle \text{PLAN} \rangle$ should show the assigned reasoning style. Do not silently upgrade to a smarter style.”

Cognitive axes (level 0 / level 2). *Contingent reasoning.* (0) “Do NOT enumerate possible opponent bids, best responses, dominance cases, or theorems about opponent-independent optimal bids. Treat the auction as a fixed environment and choose from a simple own-value rule.” (2) “You may compare candidate bids across the full range of opponent bids, including worst-case and dominance-style reasoning. If one bid is robust

across all cases, use it.” *Forward planning*. (0) “Do NOT trace through what happens after different candidate bids. Do NOT compare expected profits. Use a margin-of-safety heuristic: bid below your value...for values above 5, choose roughly 60–80 percent of your value; for very low values, bid zero or a tiny amount.” (2) “Build a decision tree over several candidate bids. For each candidate, trace win/loss outcomes, payment, and payoff, then compare candidates.” *Beliefs about others*. (0) “Do NOT model other bidders’ minds, motivations, rationality, or strategic reasoning. Do not assume they will bid truthfully or optimize. Treat them as an unknown background.” (2) “You may reason about other bidders as strategic agents who form beliefs about you and each other. Use second-order strategic reasoning when choosing a bid.” Level-1 paragraphs are intermediate (enumerate a few cases / one-step lookahead / first-order beliefs).

Error mouse 1: second-price overgeneralizer. “You strongly focus on the fact that the winner pays the second-highest bid, not their own bid. Overgeneralize this rule: treat bidding above your value as mostly safe because you expect not to pay your own bid.” Construction rationale: encode one mistaken inference (payment-rule overgeneralization), keep the rest minimal so the resulting behavior is mechanistically attributable to that single premise; the strain’s behavior is harmless in second-price and loss-making in first-price, which is the falsifiable consequence used in Section 8.

Error mouse 2: payment panic. “You are afraid that your bid is the amount you might have to pay...choose the smallest positive bid allowed by the auction grid...Even if your value is high, protect yourself by bidding 0.1 rather than trying to win.” Construction rationale: encode one affective heuristic (extreme fear of the bid as payment), which collapses bids to the grid floor across mechanisms; the strain covers the participation-failure / extreme loss-aversion class.

Round-2 error mouse 1: win_seeker. “You really want to win this auction. The reward of winning—having the prize—weighs heavily in your mind, and the cost of paying a little above what the prize is worth feels small in comparison. You bid aggressively to maximize your chance of winning, and you are willing to overbid above your value. Choose a bid somewhere between 110% and 130% of your value.” Construction rationale: a mechanism-agnostic overbidder—no rationalization via second-price rule, no rule-specific vocabulary—so the strain transfers across mechanisms without changing meaning. Used in the cluster panel (Section 9.1) and in moment matching (Section 9.3).

Round-2 error mouse 2: loss_averse. “You are a loss-averse bidder. The pain of overpaying for a prize feels much larger to you than the equivalent pleasure of winning a profitable auction. You shade your bid below your value. Choose a bid somewhere between 50% and 80% of your value.” Construction rationale: encode a Kahneman–Tversky-style loss-aversion frame as a moderate- shading class without collapsing to the grid floor. Replaces payment panic for moment matching against human distributions where near-zero bidding is rare.

B Outcome-Instruction Alternative Prompts

The procedure-instructed CoT panel of Section 4 restricts the model’s deliberation by both a cognitive-style block and a worked example, then lets the model decide a bid. A natural alternative is to restrict only the *outcome* of deliberation: tell the model what it does, without saying how to think about it. Comparing the two designs separates two explanations for the observed behavior at $c=2$ (truthful) and $c=0$ (underbid): genuine procedural induction versus simple compliance with a labeled behavior.

B.1 Construction

We build three outcome-instructed strains, each identical except for one behavior line. The persona uses the same “Digital Mouse” framing as the procedural CoT panel so the comparison is on the deficit-induction mechanism only, not on prompt style. The persona also retains the $\langle \text{PLAN} \rangle / \langle \text{ACTION} \rangle$ output format and one worked example showing the labeled behavior at two value levels.

- `spsb_outcome_overbid`: “You bid above your value.”
- `spsb_outcome_underbid`: “You bid below your value.”
- `spsb_outcome_truthful`: “You bid equal to your value.”

Each strain is run for 30 reps against the same SPSB IPV environment (`gpt-5.4-mini`, temperature 0.5, 3 bidders, private values uniform on $[0, 49]$, 0.1 grid), yielding 90 bidder-decisions per outcome strain.

Table 11: Outcome-instructed strains versus procedure-instructed CoT corners. The outcome strains comply with the labeled behavior in 100% of decisions; the CoT corners match 100% at $c=2$ (truthful) and produce a margin-of-safety underbid at $c=0$ (truthful rate ≈ 0 , ratio ≈ 0.70). Same SPSB IPV environment for both panels.

Strain	Source	n	mean b/v	truthful	target hit
spsb_outcome_overbid	outcome (alt)	90	1.01	0.000	1.000
spsb_outcome_underbid	outcome (alt)	90	0.84	0.000	1.000
spsb_outcome_truthful	outcome (alt)	90	1.00	1.000	1.000
c0_f0_b0	CoT (procedure)	90	0.71	0.056	—
c2_f2_b2	CoT (procedure)	90	1.00	1.000	—

Notes. “Target hit” is the fraction of bids in the labeled direction (above value for overbid, below for underbid, within 0.05 for truthful). The outcome-overbid strain’s mean ratio is only 1.01 because the model chooses the smallest above-value bid that complies with the grid; this is below the extreme-overbid band ($b - v \geq 5$).

B.2 Results

Table 11 reports the outcome-instructed strains’ behavior against the CoT-procedural targets at $c=0$ and $c=2$. The outcome strains are perfectly compliant: 100% of bids fall in the labeled direction.

B.3 What the comparison shows

Three observations follow.

First, the outcome-instructed strain for truthful behavior is indistinguishable from the procedure-instructed CoT $c=2$ strain at the behavioral level: both bid value 100% of the time (Table 11). At the reasoning-chain level they differ sharply: the outcome strain’s plan reads “*My behavior is to bid equal to my value*” and the procedure strain’s plan reads “*In a second-price auction... bidding my value protects both cases.*” The two strains map to the same bid, but they expose *different* cognitive paths. For a phase-I screen this matters: a mechanism that handles a truthful bidder via the dominance argument may not handle one whose truthfulness is a labeled habit, and the outcome strain is the right stressor for the latter case.

Second, the outcome-instructed overbid strain produces overbidding at 100% but with a small margin (median $b/v \approx 1.01$). This is the minimal compliance with the label: the model bids one grid step above value rather than committing to a substantial overbid. The v1 second-price-overgeneralizer strain, in contrast, bids well above value with median $b/v \approx 1.15$ in FPSB and an extreme-overbid rate of 0.44 in SPSB. The difference quantifies

what a *content* error (the overgeneralizer’s belief that “I won’t pay my own bid”) buys over a bare outcome label: more aggressive, more mechanistically interpretable overbidding. The outcome strain covers the overbidding class for support purposes but does not substitute for the content-error mouse when the screen needs an overbidder whose mistake is grounded in a specific wrong belief.

Third, the outcome-underbid strain produces a b/v ratio of 0.84, which is above the CoT-procedural $c=0$ ratio of 0.71. The procedural strain underbids more aggressively because its worked example shows a margin-of-safety bid of $\sim 70\%$ of value; the outcome strain just says “below your value” and the model takes that to mean any amount below. Outcome instructions are easier to satisfy and produce milder behavior; procedural induction with a demonstration produces a sharper, more identifiable signature. For the moment-matching step of Section 3, where the panel weights are fit to match a human moment vector, this matters: a panel that contains only outcome strains for underbidding cannot reproduce the human moment for moderate-to-strong underbidding, while a panel that contains the procedural $c=0$ strain can.

The summary message is that procedure-instructed and outcome-instructed strains are complementary, not substitutable: the former produce a sharper behavioral signature and an internal reasoning chain to audit; the latter are minimal-compliance covers that quickly span the support of the mistake space. Both belong in the library, and the choice between them is a question about *what the panel is being asked to do* at a given stage of the pipeline.

References

- Almaatouq, A., T. L. Griffiths, J. W. Suchow, M. E. Whiting, J. Evans, and D. J. Watts (2024). “Beyond Playing 20 Questions with Nature: Integrative Experiment Design in the Social and Behavioral Sciences.” *Behavioral and Brain Sciences*, 47, e33.
- Alsobay, M., D. G. Rand, D. J. Watts, and A. Almaatouq (2025). “Integrative Experiments Identify How Punishment Impacts Welfare in Public Goods Games.” arXiv:2508.17151.
- Crawford, V. P., and N. Iriberri (2007). “Level- k Auctions: Can a Nonequilibrium Model of Strategic Thinking Explain the Winner’s Curse and Overbidding in Private-Value Auctions?” *Econometrica*, 75(6), 1721–1770.
- Eyster, E., and M. Rabin (2005). “Cursed Equilibrium.” *Econometrica*, 73(5), 1623–1672.
- Horton, J. J. (2023). “Large Language Models as Simulated Economic Agents: What Can We Learn from Homo Silicus?” NBER Working Paper.

- Kagel, J. H., and D. Levin (1993). “Independent Private Value Auctions: Bidder Behaviour in First-, Second- and Third-Price Auctions with Varying Numbers of Bidders.” *The Economic Journal*, 103(419), 868–879.
- Li, S. (2017). “Obviously Strategy-Proof Mechanisms.” *American Economic Review*, 107(11), 3257–3287.
- Li, S. (2024). “Designing Simple Mechanisms.” *Journal of Economic Perspectives*, 38(4), 175–192.
- Manning, B. S., K. Zhu, and J. J. Horton (2024). “Automated Social Science: Language Models as Scientist and Subjects.” NBER Working Paper.
- Peng, T., G. Gui, D. Merlau, et al. (2025). “Digital Twins as Funhouse Mirrors: Five Key Distortions.” arXiv:2509.19088.
- Zhu, K., and coauthors (2024). “LLM Auction: Comparing Human and Language- Model Bidders across Auction Formats.” Working paper.
- Vickrey, W. (1961). “Counterspeculation, Auctions, and Competitive Sealed Tenders.” *The Journal of Finance*, 16(1), 8–37.